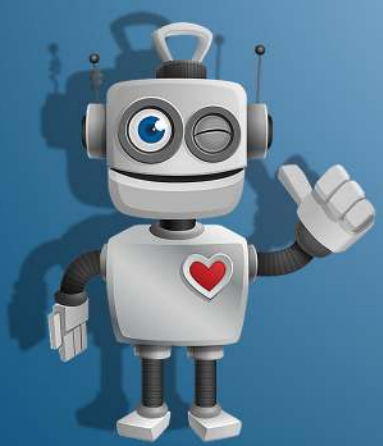
تقدم لكم اللجنة الاكاديمية في فريق

Mechatronics spectrum

اسئلة سنوات لمادة

نظرية التحكم





ميد



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1	2	3	4	5	6	7	8	9	10	11	12	13	14
B	A	D	D	C	A	B	A	A	B	A	C	D	D

1. Let $F(s)=\frac{1}{s(s+1)}$, The final value of $f(t)$ is: $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s) = \lim_{s\to 0} \frac{s}{s(s+1)}$

   a) 0 b) ∞ c) 1 d) None

2. $y(t)=e^{-t}-e^{+t}$, Then Y(s) will be

   A. $\frac{1}{s^2+1}$ B. $\frac{-2}{s^2-1}$ C. $\frac{+2}{s^2-1}$ D. None

3. Open-loop control can be used, in practice, only if:
   A. the relationship between the input and output is known B. there are no internal disturbances.
   C. there are no external disturbances D. all above

4. The transfer function of an L C series circuit, where $v_c(t)$ is the output and the applied voltage v (t) is the input, is

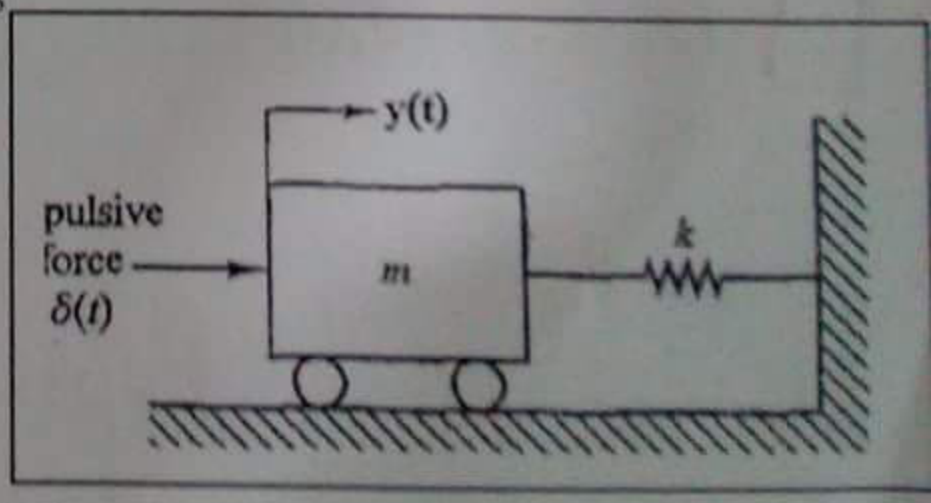
   A. $\frac{L}{LCs+1}$ B. $\frac{1}{LCs^2+1}$ C. $\frac{Cs}{LCs^2+1}$ D. $\frac{1}{LCs+1}$

5. The number of state equations in sprig-mass-dashpot linear mechanical system is
   A.4 B.3 C.2 D. 1

6. The output of the main comparator in a feedback control system is called
   A. error signal B. Feedback signal C. output signal. D. Difference signal

7. The major disadvantage of a feedback system may be
   A. Inaccuracy B. instability C. Unreliability D. inefficiency

8. For the system shown, the mathematical model relating
   The output ($y(t)$) and the input ($\delta(t)$) is
   A. $my''(t)+fy'(t)+Ky(t)=\delta(t)$
   B. $my''(t)+Ky(t)=\delta(t)$
   C. $my''(t)+fy'(t)+Ky(t)=0$
   D. $my''(t)+Ky(t)=0$



9. The state variables of a system given by $y'''(t)+5y''(t)+4y'(t)+2y(t)=3u(t)$ are
   A. $x_1(t)=y(t); x_2(t)=y'(t); x_3(t)=y''(t)$ B. $x_1(t)=2y(t); x_2(t)=4y'(t); x_3(t)=5y''(t)$
   C. $x_1(t)=y(t); x_2(t)=r(t)$ D. $x_1(t)=y(t); x_2(t)=y''(t)$

10. The steady state error in the unit ramp response of first order standard system is

    A.0 B. T C 2T D. 3T

MECHATRONICS SPECTRUM TEAM

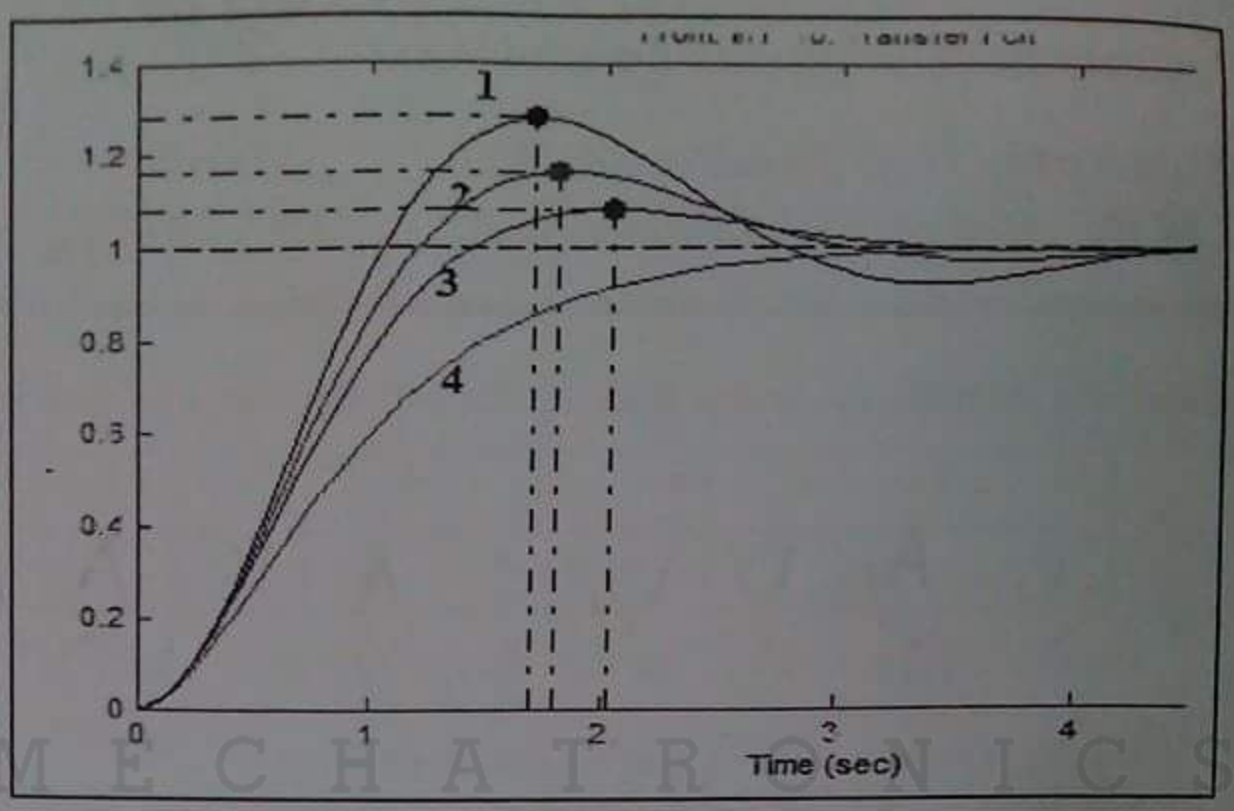
Depend on the following figure to solve 11 & 12.

11. The time response of a second Order system given by

$$\frac{C(s)}{R(s)} = \frac{4}{s^2 + 2s + 4}$$

is denoted as

A. 1 B. 2 C. 3 D.4



12. The unacceptable response with respect to the rise time is denoted by

A. 1 B. 2 C. 3 D. 4

13. The partial fraction of $y(s) = \frac{1}{s^2(s+1)^2}$ is

A. $y(s) = \frac{1}{s^2(s+1)^2}$ B. 2 $y(s) = \frac{a}{(s+1)^2} + \frac{b}{s^2}$

C. $y(s) = \frac{\alpha_1 s + \alpha_2}{(s+1)^2} + \frac{b}{s^2}$ D. $y(s) = \frac{\alpha_1 s + \alpha_2}{(s+1)^2} + \frac{\beta_1 s + \beta_2}{s^2}$ ← ans D

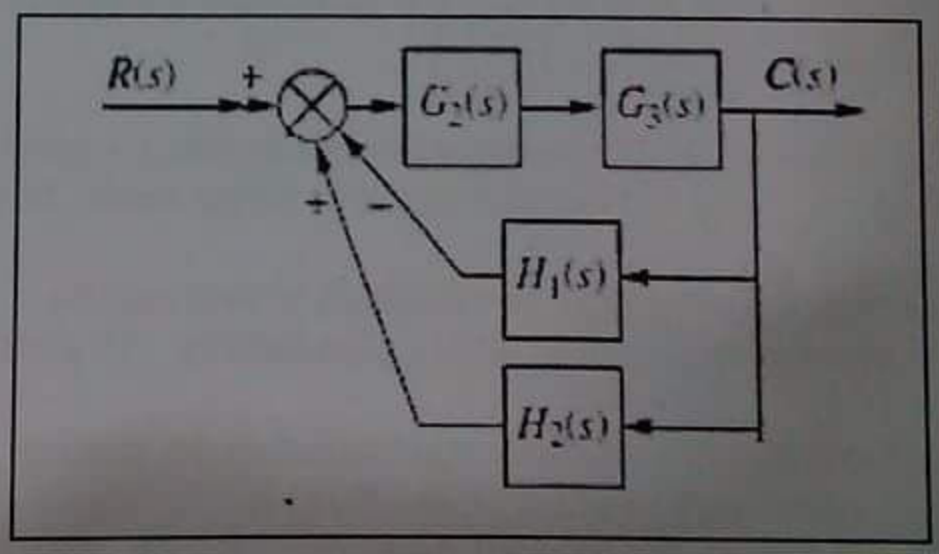
14. The open loop transfer function from the given block diagram is

A. $G_2G_3H_1H_2(s)$

B. $G_2G_3H_1(s)$

C. $G_2G_3H_2(s)$

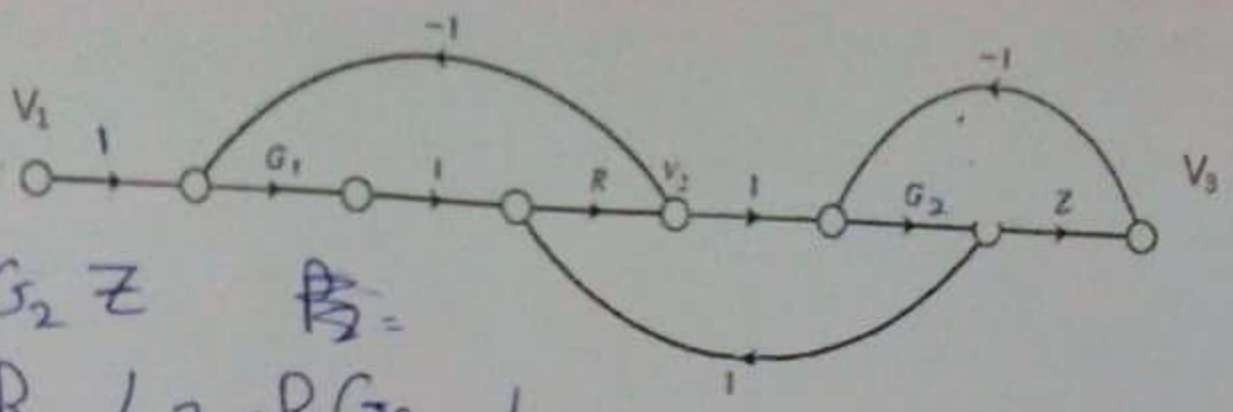
D. $G_2G_3(H_2(s) - H_1(s))$



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$P_1 = G_1 R G_2 Z$ $P_2 =$

$L_1 = -G_1 R$ $L_2 = R G_2$ $L_3 = -G_2 Z$

untouch loop

$L_1 L_2 = -G_1 G_2 R^2$ $L_1 L_3 = +G_1 G_2 R Z$ $L_2 L_3 = -R G_2^2 Z$

$\Delta = 1 - [-G_1 R_1 + R G_2 - G_2 Z] + [-G_1 G_2 R^2 + G_1 G_2 R Z - R G_2^2 Z]$

$\Delta_1 = 1$

$$\frac{V_3}{V_1} = \frac{\Delta 1 * P_1}{\Delta} = \frac{1 * G_1 R_1 G_2 Z \cdot 1}{1 - [G_1 R_1 + G_2 R_1 - G_2 Z] + [-G_1 G_2 R^2 + G_1 G_2 R Z - R G_2^2 Z]}$$

MECHATRONICS SPECTRUM

Q3) (6 marks) Assume zero initial conditions, solve the following differential equation using Laplace transform

$$y''(t) + 3y'(t) + 2y(t) = 1$$

$$s^2 y(s) + 3s\, y(s) + 2y(s) = \frac{1}{s}$$

$$y(s)\left[s^2 + 3s + 2\right] = \frac{1}{s}$$

$$y(s) = \frac{1}{s[s^2 + 3s + 2]} \Rightarrow \frac{1}{s(s+2)(s+1)} \Rightarrow \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+1}$$

$A(s+2)(s+1) + B(s+1)s + C(s+2)s = 1$

$B = \frac{1}{2}$ $A = \frac{1}{2}$ $C = -1$

$$\int \frac{\frac{1}{2}}{s} + \int_{\alpha}^{-1} \frac{1/2}{s+2} + \int_{h}^{+1} \frac{-1}{s+1}$$

$$\Rightarrow \frac{1}{2} + \frac{1}{2} e^{-2t} - e^{-t} \quad \#$$

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